

Where is all the NIH data? A Landscape Analysis of the 150 NIH Repositories and the Path Beyond the Mountains

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Abstract

The National Institutes of Health (NIH) is heavily invested in developing data repositories. To date, the NIH has funded hundreds of separate repositories, hosting exabytes of diverse data types that enable scientific research across domains, organisms and diseases. These data drive scientific discovery and power the secondary data use economy guided by principles of findability, accessibility, interoperability and reusability (FAIR). While this explosion in data resources should in theory make more data available for research, it also scales new operational challenges. Looking beyond the FAIRness of the data hosted in the repositories, we must examine the FAIRness of the repositories themselves. In this paper, we report findings from landscape analysis of NIH repositories. Using a structured search approach, we inventoried 143 active NIH repositories for the types of data they contain, characterize the access mechanisms unique to each repository, and estimate their operational costs based on publicly available information on NIH Reporter. A total of 46 repositories (32.2%) hosted human genomic data, 14 repositories (9.8%) held non-human genomic data, 33 repositories (23.1%) included imaging data, and 38 repositories (26.6%) hosted clinical data. Between 1989 and 2024, the NIH has allocated an estimated \$6,718,994,379.00 for repository development, with the National Cancer Institute as the largest single-institute funder.

We instead propose that funders should prioritize funding for fewer, better connected data repositories that serve distinct purposes. Put simply, ‘*more* is more’ when it comes to generating quality datasets for rigorous biomedical data science. But ‘*less* is more’ when establishing repositories that help scientists easily find and securely access distinct data types for their research. This level of cross architecture coordination and federation will require support from both research grantors and recipients.

Introduction

The National Institutes of Health (NIH) invests millions of dollars every year to develop repositories that host the explosive growth in human clinical data, genomic data, biospecimens, and other data types for research. The NIH has funded the establishment of at least 149 repositories to date. Each repository holds data that enable scientific research across domains, organisms and diseases.¹ These data represent the scientific benefits generated from the NIH's billions of dollars of research investments, and have also been instrumental in enabling secondary research that leverages existing data. While this explosion in data resources should in theory make more data available for research, it also brings unique challenges in making sure researchers can find, access, interoperate, and reuse data per the FAIR (findable, accessible, interoperable) principles.²

In this paper, we use publicly available information from official NIH dashboards to a) inventory the types, size and costs (where available) of NIH-supported repositories hosting genomic data and b) catalog their access controls. In so doing, we aim to provide the genomics community with a resource for where to locate data, promoting at least two principles of the FAIR framework: findability and reusability. We conclude with recommendations for how both research grantors and recipients dedicated to repository development could enhance accessibility and interoperability by prioritizing efforts to stream the quality and governance of data hosted by existing repositories.

Background

Though these repositories are funded by many of NIH's Institutes and Centers (ICs) with distinct scientific foci, there is significant overlap in the features and types of data NIH repositories host. Myriad repositories host the same types of data, for the same species and diseases, with similar yet distinct mechanisms for data access. Furthermore, a disparate and disconnected NIH data repository landscape can challenge how easily and effectively researchers find, access, and re-use data housed within funded repositories. While the ability to easily federate across repositories is limited, promising developments are present in the NIH Cloud Platforms Interoperability (NCPI) program³ and in the Database of Genotypes and Phenotypes (dbGaP). NCPI is implementing the Global Alliance for Genomics and Health (GA4GH)⁴ standards in its involved repositories, including the NHGRI AnVIL,⁵ NHLBI BioData Catalyst,⁶ Gabriella Miller Kids First Data Resource Center,⁷ and the NCI Cloud Resource Data Commons (CRDC),⁸ and increasingly with the All of Us program.⁹ dbGaP is currently leveraging FHIR standards to increase the exchange of data between dbGaP and multiple, often NIH-funded, analysis platforms.¹⁰

Looking beyond the FAIRness of the data within the repositories, we must examine the FAIRness of the repositories themselves. Data repositories do not need to differ greatly in functionality to serve a majority of scientific data types, disease areas, and access paradigms. The use of tabular data and file data is ubiquitous in data science broadly. To paint an analogous example, a personal computer stores tables, images, pdfs, documents and more, using a single file-folder storage structure with little distinction needed for differences in file types or contents across the different domains of data provided to it. Similarly, cloud storage—remote data storage, which is increasingly used by NIH repositories—operates exactly the same. Meaning, a unique *repository* does not need to exist for every data type, disease area, access paradigm, species or funder.

These repositories can be difficult for researchers to discover when looking for data to advance their research. It can also be difficult for researchers to know which repositories they can use to ethically share data in compliance with the NIH DMS policy or a journal’s publication requirements.¹¹

To aid researchers in this task, and spur discussion on an easily discoverable landscape of NIH data, we inventoried all NIH repositories that currently host data, catalog the types of data contained therein, and estimate their costs. This is the first inventory to our knowledge that maps what, where, and how data stemming from NIH funded research are made available for research use. We conclude with recommendations for how to further improve on data findability in ways that respect the evolving governance and data access management policies within the NIH.

Methods

At the time of writing, an exhaustive list of 149 NIH-funded repositories was available at sharing.nih.gov. We included repositories in our analysis that enabled the collection, storage, or curation of human genomic data, non-human genomic data, imaging data, and clinical data. All data captured for and presented in this landscape analysis were publicly available via repository websites without requiring authentication or authorization. For each listed repository, we extracted the following information, when available:

- Name of repository
- Institute(s) or center(s) that provided funding
- Repository URL
- Data type(s) housed
- Access mechanism (e.g. open or controlled)
- Primary uses of the repository
- Species investigated
- Size of the data collection (e.g. petabytes, terabytes)

- Number of unique datasets or relative units (e.g. datasets, studies, participants)
- Search tools available (e.g. study, dataset catalog, participant level)
- Disease(s) investigated
- Contact information for repository manager(s) or access committee
- Date last accessed, and any additional notes related to governance structure.

While cataloging the data, we quickly realized that the repositories use different “units” in assessing the count of unique datasets within their collection. These “units” range from studies, datasets, samples, participants, compounds, files, clinical assessments, and more. Therefore, we decided to not include it in our analysis. Furthermore, few repositories listed the actual data size of their data so we determined it was insufficient to create an accurate depiction or analysis.

Of the 149 repositories listed on the NIH website, five repositories were duplicates or were inactive. One repository website returned an error. A total of 143 “active” repositories are included in our analysis (**Table 1**). Since beginning the landscape analysis in December 2024, two repositories were removed for unknown reasons, and four new repositories were added. Our comprehensive list changed accordingly. Our inventory and descriptive statistics are based on data confirmed as of July 2025.

Total Number of Repositories Included in Study	
Total Number of Repositories on NIH List	149
Total Number of Duplicate or Inactive Repositories	5
Total Number of Repositories Unable to Access	1
Total Number of Active Repositories Included	143

Table 1. Total Number of Repositories Included in Study

We also attempted to capture the total direct financial investment in each repository through analysis of related awards in NIH RePORTER.¹² Very often, multiple aims of an award and financial reports and specific allocation of costs to repository developments versus other research aims and activities are opaque. Finding the funding for repositories varied. Several listed the grants they are funded through clearly on their website. For others, the grant information was found through journal articles that the repository listed for citation reasons. In some cases, the grant information was nowhere to be found, so the name of the repository, and acronyms used, were searched in RePORTER. Oftentimes, there were hundreds, or thousands, of results, so we filtered them by information found publicly on the repository website such as: primary investigator’s name, NIH funding institution, and institution or university running the repository. Despite this, there are still repositories that funding could not be estimated. A few repositories noted a government contract, often with a third-party, such as Leidos, for the creation of their repository. However, when searching for the contracts, the funding appeared to be proprietary

information not publicly available; therefore they were counted as funding estimates “not available.” Furthermore, some repositories listed grant numbers that produced no results on RePORTER, as well as not producing results when searched by name, and were therefore excluded.

However, we only sought to include funding pertaining to the development of and maintenance of the repository. Thus, we often excluded certain “cores” of grants that were not relevant. We also excluded grants focused on specific research projects rather than the repository itself. For each repository that we managed to find funding, we included the grant number, amount of funding awarded, and fiscal year granted. Some repositories shared funding with each other, so we kept them separate and noted which repositories were included in each grant. It should be noted that, for example, the National Library of Medicine (NLM) grants were listed for “molecular biology, bioinformatics, and human genome resources” and often only listed a few of their repositories. Hence, some NLM repositories were not mentioned by name and thus we considered estimated funding as “not available.” Most repositories also receive funding from organizations outside of the NIH, such as the National Science Foundation, but we did not include this funding in our totals as we were only investigating NIH funding. We managed to estimate funding for 98 repositories (68.5%) and shared funding for 15 repositories (10.5%), yet were unable to find funding for 30 repositories (21.0%).

To gather the total funding by year, we aggregated all the found data from each repository and the shared repositories and sorted it by year. Our data ranged from 1989-2025, however, we excluded the 2025 data from our results as the full fiscal year is not reflected yet. The total NIH funding to research grants ranges from 1998-2024 and can be found on RePORTER’s Data Book.^{13,14}

Results

Of the 143 repositories included in our study, 87 (60.8%) are open access, 31 (21.7%) are controlled access, and 25 (17.5%) have a combination of open and controlled access, depending on the data requested (**Figure 1**).

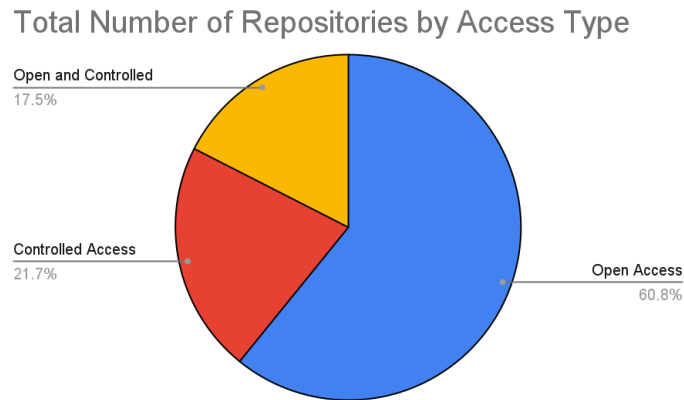


Figure 1. Total number of repositories by access type

A total of 46 repositories (32.2%) hosted human genomic data, 14 repositories (9.8%) held non-human genomic data, 33 repositories (23.1%) included imaging data, and 38 repositories (26.6%) hosted clinical data. It should be noted that some repositories house several kinds of data, and others may house alternative or additional data types that were not included in these results (**Figure 2**).

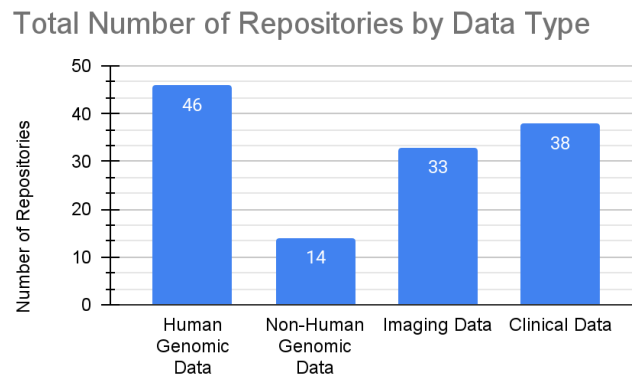


Figure 2. Total number of repositories by data type

Of the data hosted by the 143 active repositories, humans (83.2%) are the most prevalent species investigated, followed by mice (21.7%), and rats (11.2%), as seen in **Figure 3**. Some repositories house data from multiple species. It should also be noted that seven repositories had data from such a wide range of species that we simply noted it as “wide variety.”

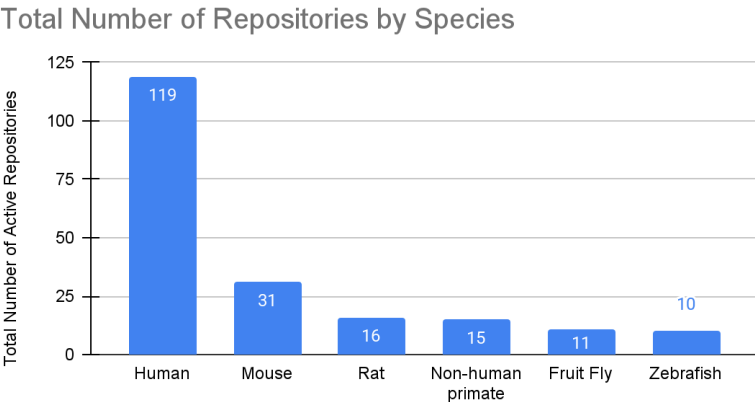


Table 3. Total Number of Repositories by Species

NIH repositories are funded by the various NIH institutes and centers. According to the NIH list, 113 repositories (79.0%) are sponsored by one institute. Another 30 repositories (21.0%) are sponsored by multiple institutes. The National Cancer Institute (NCI) is the largest single-institute funder for 23 repositories (16.1%), the National Institute on Aging (NIA) with 22 repositories (15.4%), and the National Institute of Neurological Disorders and Stroke (NINDS) with 19 repositories (13.3%). **Figure 4** displays the number of repositories sponsored by each institute and the types of access.

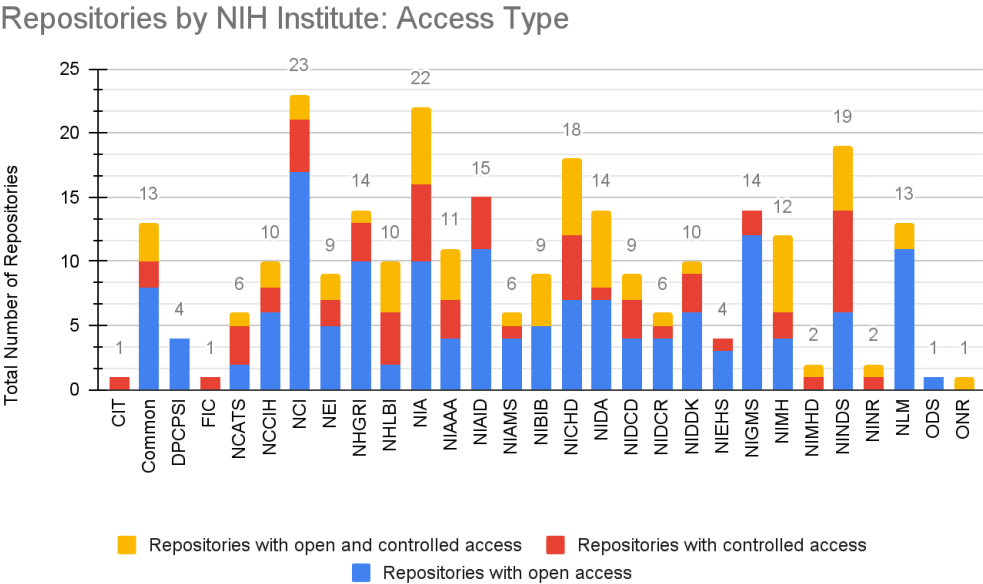


Figure 4. Repositories by NIH institute: access type and total number of repositories

Figure 5 graphs the types of repository data by sponsoring institute. Some repositories house several types of data, causing the total numbers to appear to be larger than the actual total number of repositories.

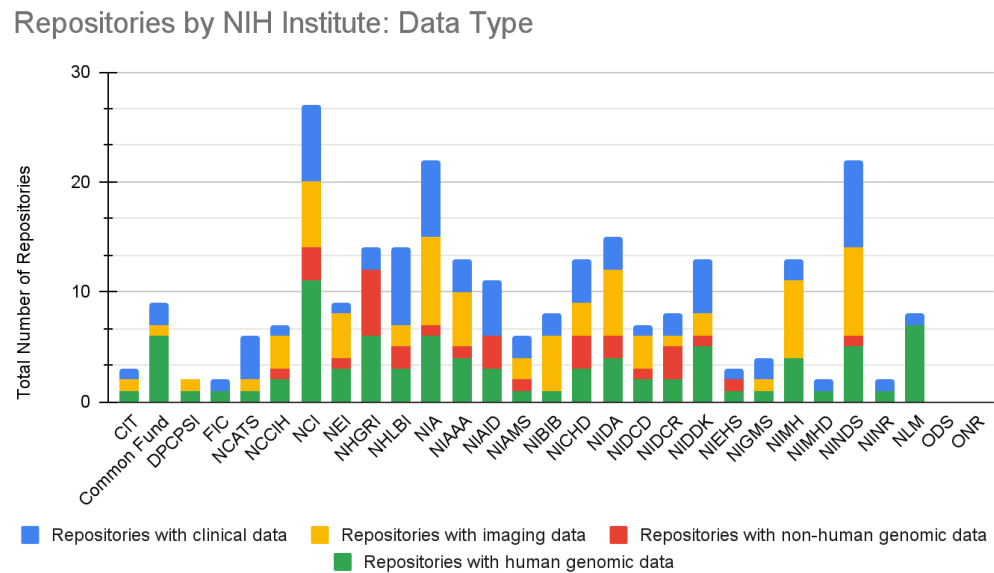


Figure 5. Repositories by NIH institute: Data Type

Several “disease categories” are covered by multiple repositories (**Figure 6, 7**). Notably, 33 repositories (23.1%) focused on neurological disorders and neurology, 23 repositories (16.1%) investigated cancer, and 15 repositories (10.5%) were dedicated to infectious diseases. An estimated 56 repositories (39.2%) did not focus on one particular disease, of which, 44 repositories enabled access to data of other scientific types (e.g., anatomy, chemical banks, etc.).

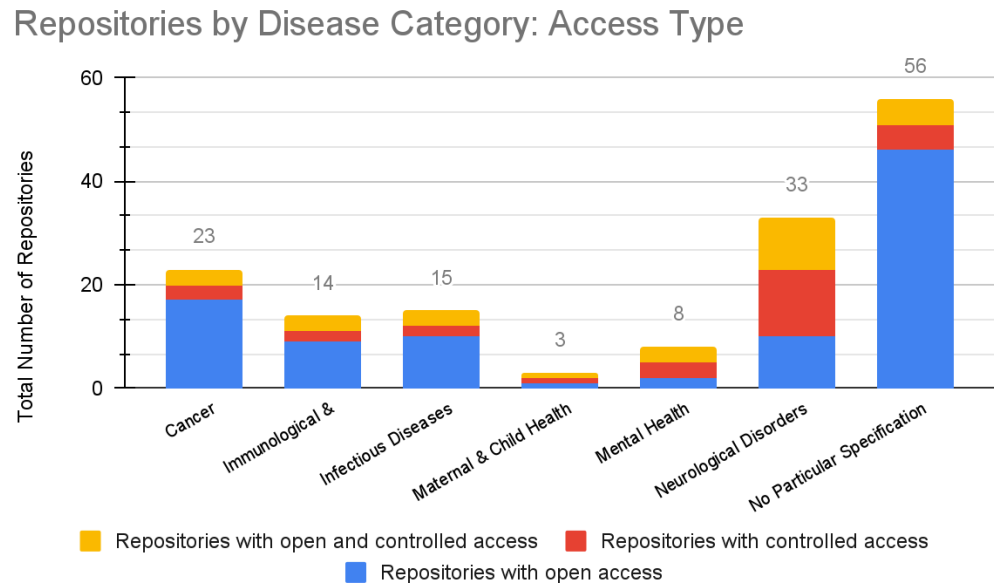


Figure 6. Repositories by disease category: access type and total number of repositories

Repositories by Disease Category: Data Type

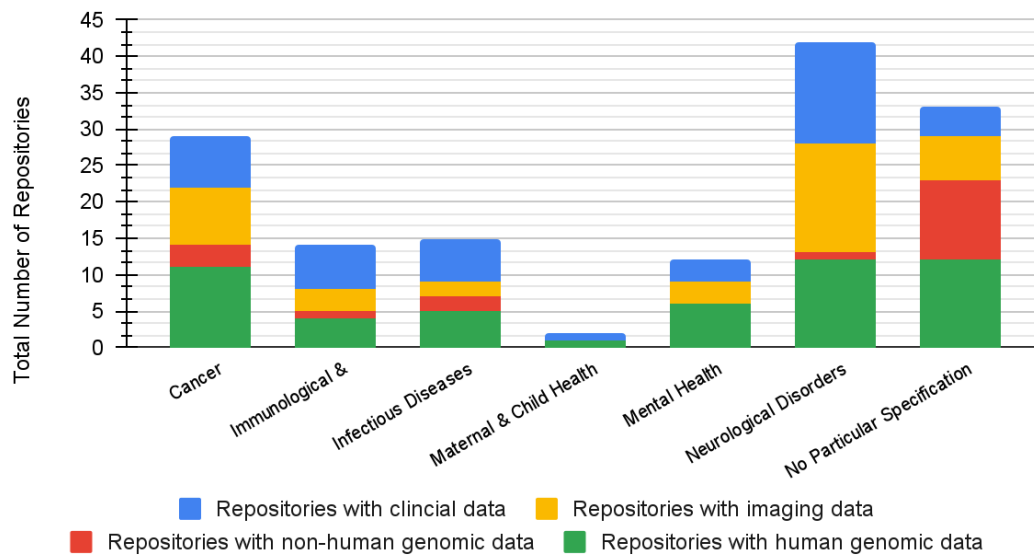


Figure 7. Repositories by disease category: data type

Within each disease category, we also noted specific diseases that appeared to be frequently investigated. Among these are Alzheimer's, COVID-19, and Parkinson's, each with 8 repositories (5.6%). The total repositories and access types of specific, frequently investigated diseases can be seen in **Figure 8** and the types of data in **Figure 9**.

Repositories by Disease: Access Type

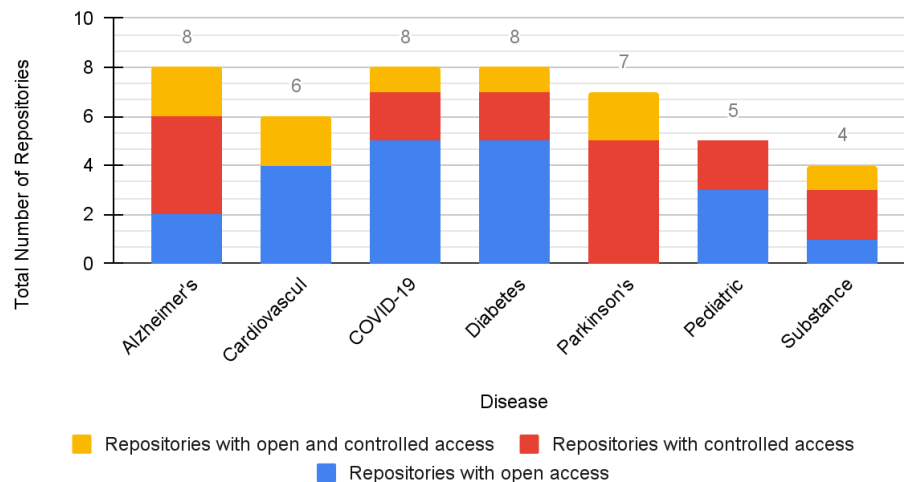


Figure 8. Repositories by disease: access type and total number of repositories

Repositories by Disease: Data Type

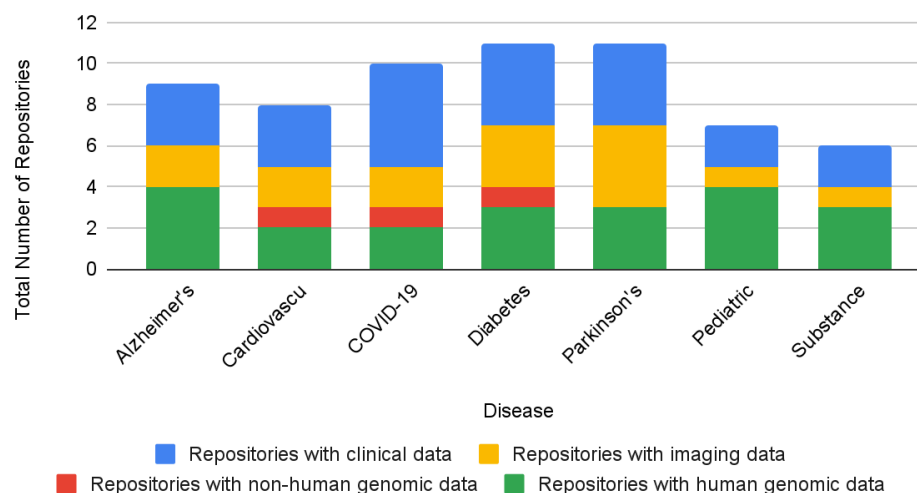


Figure 9. Repositories by disease: data type

The total estimated funding repositories received from the NIH between 1989-2024 is \$6,718,994,379.00. However, because we were unable to identify funding from 30 repositories and due to our exclusion criteria, this can be assumed to be a very conservative estimate. Compared to the total NIH funding to research grants from 1998-2024, this estimated sum is 1.5% of the spending (**Figure 10**).

Estimated Repository Funding vs. Total NIH Funding

1989-2024

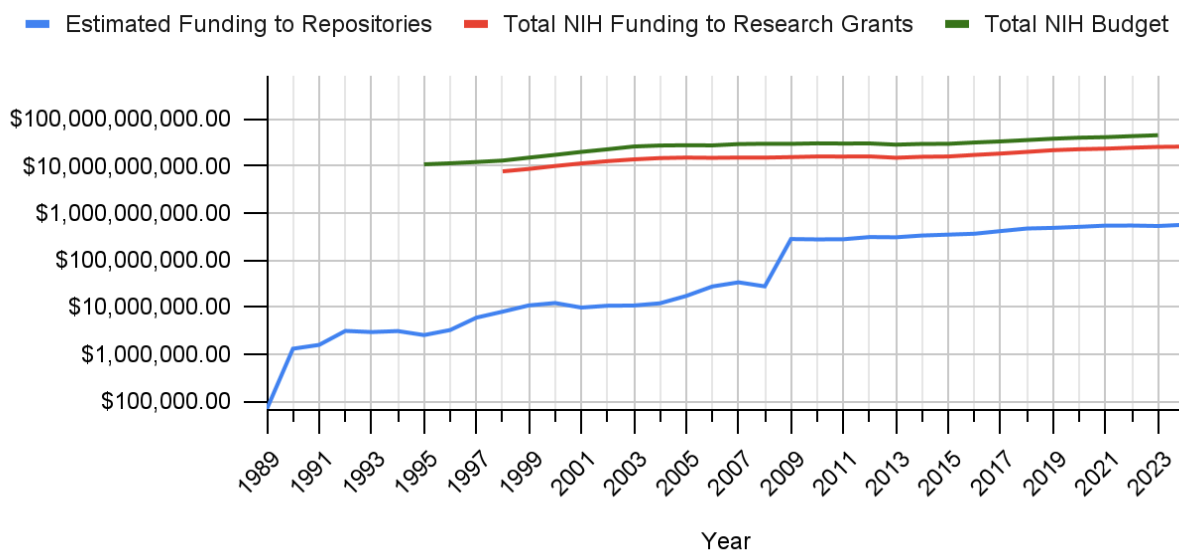


Figure 10. Estimated Funding to Repositories vs. Total NIH Funding to Research Grants vs. Total NIH Budget (1998-2024)

Discussion

Counting the cost

Efforts to establish these 149 data repositories did not occur overnight, but rather over decades of sustained investment in research data infrastructure that has enabled the sharing of scientific data and results to advance science, cure disease, and alleviate human suffering. In addition to research skills, demand has been steadily growing for data curation and database design required to securely maintain repository infrastructures. Research funding has had to account for these additional skillsets and operational needs, which are reflected in the upward trendlines for total funding for repository development.

Maintaining a broad array of repositories, some hosting similar or identical data types, will continue to create a high cost burden. Each of these repositories possess technologies that deprecate, require updating, and expose security vulnerabilities that must be remediated. The scientific and research funding communities together face a new challenge related to addressing the issue not of too few databases, but of too many. Integrating across or consolidating repositories, especially those that are functionally similar, is beneficial for scientists, funders, as well as those who altruistically contributed their data to research in the first place. It supports the FAIRification of datasets in that similar data types can be more easily located by researchers given there are fewer places to look; can be accessed by authorized users with more streamlined access controls; can be federated with other data types to enhance analytic power; and can be reused to reduce redundancy in data collection and curation.

Without bridging repository architecture, funders can expect to multiply the work of repository a) maintenance and operations, b) security reviews and updates, c) search tool development, d) data engineering (ingest and lifecycle management), and e) access management without the added benefit of more data or science.

Where else can we go?

In highlighting the need for federated or consolidated architecture efforts, we do not suggest an ‘optimal data infrastructure.’ We instead argue that funders should prioritize funding for fewer connected data repositories that serve distinct purposes. Put simply, ‘*more* is more’ when it comes to generating quality datasets for rigorous biomedical data science. But ‘*less* is more’ when establishing repositories that help scientists easily find and securely access distinct data types for their research. This level of architecture federation or consolidation will require support from both research grantors and recipients. Fortunately, there are several bright spots of integration aided by groups like dbGaP and the NCPI platforms that can serve as helpful use cases.

While data repositories may confidently remain agnostic to data types, the search tools needed to explore them may instigate a need for more custom software to surface and sort through distinct

data elements. Yet the success demonstrated by the Computational Genomics Lab (CGL) at UCSC's Data Explorer used in the AnVIL,¹⁵ Human Cell Atlas,¹⁶ and LungMAP¹⁷ programs, along with the Broad Institute's Terra Data Explorer which has been applied to multiple distinct datasets with only weeks of tailoring, shows that dataset-specific exploration at the participant-level can be achieved, and perhaps most efficiently if needed repeatedly, by using a common software infrastructure that can be tailored to specific datasets with minor modifications.

Converging Data Repositories, Seeding Diverse Analysis Tools

The stabilizing and convergent nature of data repositories stands in stark contrast to the rapidly diverging possibilities and needs for data analysis platforms and tools. Though data repositories are not without ongoing needs and opportunities, novel analysis capabilities are more uniquely tied to data types and scientific domains and their evolution, and likely warrant greater investment of research and development funds.

NIH would do well to continue funding researchers incubating these novel analysis products through traditional research funding while supporting stable and secure maintenance of existing data repository infrastructure via other funding sources, such as traditional vendor agreements and software licenses. As our landscape analysis shows, NIH's historic use of grant funding to germinate novel data sharing methods, for example to establish cloud-based data repositories and analysis ecosystems, has been very successful. Although necessary, many of these repository elements are no longer novel. Moreover, while grant mechanisms generally serve novel research and development, their regular expiration and detailed progress and financial reporting requirements make them a poor fit for long-running infrastructure needs. This can put the long term sustainability of each repository at risk. Traditional licensing models, especially those with negotiated, long term rates to protect against price gouging or volatility in the market, can lower the overhead costs of repository maintenance while likely affording NIH more control over software contracts.

Toward a Hub & Spoke Model

If the NIH biomedical software ecosystem was allowed to mature as proposed, the research community would imply consolidation to a set of data repositories that could serve as a centralized and federated *data hub* that more proximately co-located data, standardized and facilitated data submission, search, and access, and in the process likely reduce overall NIH spending on repositories through lowered maintenance and security costs. This consolidation and co-location would also breed more easily obtainable opportunities for current and future analysis tools, such as those invested in through the course of NIH research grants, to qualify themselves as meeting the appropriate NIH security standards to analyze NIH (at times, controlled-access) data.¹⁸ A secure 'plug-and-play' architecture that allows data to be sent to the *secure analysis*

tool spokes could accelerate time-to-discovery that may be limited by the current repository landscape.

Conclusion

The community of NIH-funded scientists has experimented and iterated on the ideal data repository structures and search tools. The significant overlap in the functional uses of these repositories demonstrates general agreement on the infrastructure necessary to support scientific data sharing. Common ideal capabilities of these repositories often includes data ingestion, egress, versioning, access request and provisioning, sufficient security compliance, and increasingly cloud storage capabilities which enable multiple users to work from a single copy - which helps reduce the data storage cost nested in NIH research spending. As a result, we have a growing set of repositories with converging functionalities. Meaning, NIH is paying to implement similar software multiple times over. While ‘casting the bread upon many waters’ was likely a wise way to foray into software enabled data sharing in the advent of cloud computing, many of the necessary early experiential lessons have been learned and the significance of the investment and the number of repositories can be reduced without losing value.

Further, and perhaps counterintuitively, consolidating repositories will likely benefit data sharing and expedite research discovery by making the data more accessible and readily available for co-analysis due to co-location and systems consolidation with mature integrations. In addition to the use-benefits of a consolidated repository architecture for NIH data, this will likely also liberate research dollars to fund next-generation analytic tools that will drive the next genomic revolution.

Our proposal, simply put, is that consolidating repositories will optimize our nation’s scientific data infrastructure while saving money that can be put toward more innovative research and development.

Supplementary Materials

NIH List of Repositories - Downloaded July 29, 2025 from:

<https://sharing.nih.gov/data-management-and-sharing-policy/sharing-scientific-data/repositories-for-sharing-scientific-data>

Appendix of Tables and Data

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